electronics

Article

FairChain: A Trusted and Transparent Blockchain-Based Ecosystem for Drug Development for Nagoya Protocol Implementation

Shada AlSalamah [1,*], Shaima A. Alnehmi [1], Anfal A. Abanumai [1], Asmaa H. Alnashri [1], Sara S. Alduhim [1], Norah A. Alnamlah [1], Khulood AlGhamdi [2], Haytham A. Sheerah [3], Sara A. Alsalamah [4,5] and Hessah A. Alsalamah [1,6,*]

1 College of Computer and Information Sciences, King Saud University, Riyadh 11495, Saudi Arabia
2 King Saud University Medical City, Riyadh 11495, Saudi Arabia
3 The Advisory Unit, Ministry of Health, Riyadh 11451, Saudi Arabia; hasheerah@moh.gov.sa
4 Department of Computer Science, Virginia Tech, Alexandria, VA 22305, USA; salsalamah@vt.edu
5 College of Computer and Information Sciences, Imam Mohammad Ibn Saud Islamic University, Riyadh 11623, Saudi Arabia
6 College of Engineering, Al Yamamah University, Riyadh 13541, Saudi Arabia
* Correspondence: saalsalamah@ksu.edu.sa (S.A.); halsalamah@ksu.edu.sa (H.A.A.)

Abstract

The coronavirus pandemic has spread globally, affecting over 700 million people and resulting in over 7 million deaths. In response, global pharmaceutical companies and disease control centers have urgently sought effective treatments and vaccines. However, the rise of counterfeit drugs has become a significant concern amid this urgency. To standardize the legal provision and usage of genetic resources, the United Nations Development Program (UNDP) introduced the Nagoya Protocol. Despite advancements in drug research, the production process remains tedious, complex and vulnerable to fraud. *FairChain* addresses this pressing challenge by creating a transparent ecosystem that builds trust among all stakeholders throughout the Drug Development Life Cycle (DDLC) by using decentralized, immutable, and transparent blockchain technology. This makes *FairChain* the first digital health tool to implement the principles of the UNDP's Nagoya Protocol among all stakeholders throughout all DDLC stages, starting with sample collection, to discovery and development, to preclinical research, to clinical development, to regulator review, and ending with post-market monitoring. Therefore, *FairChain* allows pharmaceutical companies to document the entire drug production process, landowners to monitor bio-samples from their land, doctors to share clinical research, and regulatory agencies such as the Food and Drug Authority to oversee samples and authorize production. *FairChain* should enhance transparency, foster trust and efficiency, and ensure a fair and traceable DDLC. To date, no blockchain-based framework has addressed the integration of traceability, auditability, and Nagoya Protocol compliance within a unified system architecture. This paper introduces *FairChain*, a system that formalizes these requirements in a modular, policy-aligned, and verifiable digital trust infrastructure.

Keywords: blockchain technology; drug development; pharmaceutical; digital health; Nagoya protocol

check for updates

Academic Editor: Aryya Gangopadhyay

Received: 14 April 2025
Revised: 27 May 2025
Accepted: 30 May 2025
Published: 22 June 2025

Citation: AlSalamah, S.; Alnehmi, S.A.; Abanumai, A.A.; Alnashri, A.H.; Alduhim, S.S.; Alnamlah, N.A.; AlGhamdi, K.; Sheerah, H.A.; Alsalamah, S.A.; Alsalamah, H.A. *FairChain*: A Trusted and Transparent Blockchain-Based Ecosystem for Drug Development for Nagoya Protocol Implementation. *Electronics* **2025**, *14*, 2527. https://doi.org/10.3390/electronics14132527

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

The advent of novel pharmaceuticals has significantly influenced global health metrics, enhancing life expectancy by 0.75–1.0% and contributing to a corresponding increase in lifetime income [1]. Research has consistently demonstrated a strong correlation between drug innovation and an upward trend in life expectancy. The 33 percent increase in life expectancy in the US between 1990 and 2015 is attributed in part to drug innovation, among other factors such as public health and other (non-pharmaceutical) developments in medical care [2]. Notably, a lack of ongoing pharmaceutical advancements could potentially not only halt this progress but perhaps even reverse it [1,2].

The journey of bringing new pharmaceutical solutions from conception to market is described by the Drug Development Life Cycle (DDLC) [3]. This meticulous process comprises six critical stages: sample collection, discovery and development [4], preclinical research, clinical development, regulator review, and post-market monitoring [5] (see Figure 1).

When exploring natural drug sources, researchers encounter the field of pharmacognosy. This discipline investigates plants and other organic materials, known as bio-samples, in the quest for new therapeutic agents [6]. Our planet hosts a rich variety of these bio-samples, many of which are already employed in medicine, while a vast reservoir also remains untapped. Local communities often possess valuable knowledge regarding the medicinal applications of these bio-samples, offering a wealth of potential genetic solutions that could transform global healthcare. This path from collection to commercialization includes three core steps in the discovery phase: sample collection, purification, and rigorous screening [3,4].

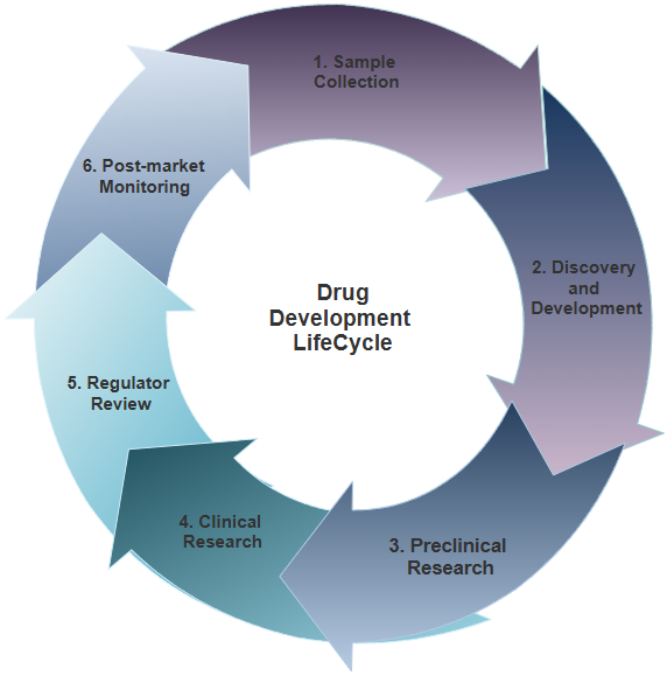


Figure 1. The Drug Development Life Cycle (DDLC) illustrates the six major stages in modern pharmaceutical development, starting with bio-sample collection and ending with post-market safety monitoring. This serves as the foundational process that *FairChain* aims to support and document using blockchain technology.

However, the complex relationships among researchers, drug developers, and source countries add another layer of complexity to the process [7]. The extraction and export of these biological resources requires strict compliance with regulations in order to foster a mutually beneficial and ethically balanced relationship [7]. The Fair Access and Benefit-Sharing Protocol facilitates this balance, ensuring that communities providing bio-samples

receive appropriate compensation [7]. Whether this compensation is monetary, such as direct payments or profit shares, or non-monetary, e.g., knowledge sharing or skill development, depends on the commercial trajectory of the bio-sample, and particularly on whether it evolves into a widely marketable pharmaceutical product.

The acquisition of bio-samples by international researchers or pharmaceutical companies requires explicit permission from the host nation [7], while the global export of these biological specimens mandates official authorization [7]. The Fair Access and Benefit-Sharing Protocol was established to ensure that countries and local stakeholders providing these bio-samples receive fair compensation from their commercial applications [7]. Whether in monetary form (e.g., direct payments or profit shares) or non-monetary benefits (e.g., technology transfers or training), this compensation is especially crucial when bio-samples evolve into marketable pharmaceutical products [7].

In recognizing the essential role of the original landowners in discovering new drug components, pharmaceutical companies must ensure that these contributors are duly compensated. Given the lengthy drug development process which spans approximately 8–12 years [8], omitting the contributions of these landowners risks undermining their rightful recognition and leading to inequities in profit distribution. Therefore, in 2010 the United Nations Development Program (UNDP) introduced the Nagoya Protocol, a transparent legal framework to support the Convention on Biological Diversity Goals. These goals focus on access to genetic resources and the equitable distribution of benefits derived from them [9]. As a consequence, the Nagoya Protocol offers a foundational framework for documenting and protecting the rights of original landowners throughout the drug development process.

FairChain aims to chronicle the life cycle of bio-samples and meticulously document information about the original landowners, thereby safeguarding their rights. A cornerstone of *FairChain* is its integration of blockchain technology. With the inherent attributes of immutability and traceability, blockchain provides an ideal platform to secure the vision put forward by the Nagoya Protocol [10], ensuring balanced benefits for original landowners, researchers, pharmaceutical companies, and the global community. The choice of blockchain over conventional security architectures such as centralized databases or private cloud-based access control systems was driven by several key factors. First, blockchain offers a tamper-resistant ledger that ensures the integrity and non-repudiation of transactions, which is essential for maintaining trust in multi-stakeholder environments such as pharmaceutical ecosystems. Second, its decentralized architecture mitigates the risks associated with single points of failure that are common in centralized systems. Third, blockchain's built-in support for audit trails via smart contracts ensures transparent enforcement of data policies, which aligns with the traceability requirements of the Nagoya Protocol. Compared to other security solutions, blockchain uniquely combines cryptographic integrity, operational transparency, and distributed trust without requiring intermediaries.

Furthermore, the *FairChain* platform enables a seamless record of each stage of the DDLC, from initial bio-sample registration to the final approval phase. Leveraging blockchain's capabilities, *FairChain* functions as a decentralized digital ledger adhering to the highest standards of data security and confidentiality [11]. Therefore, *FairChain* facilitates the democratic dissemination of bio-sample ownership data to various global stakeholders across the DDLC. Notably, *FairChain* is set to become the leading solution supporting the effective implementation of the NP.

The novelty of *FairChain* lies in its integration of the Nagoya Protocol's Access and Benefit-Sharing (ABS) mandates into an operational and verifiable blockchain-based system that spans the entire Drug Development Life Cycle (DDLC). Unlike previous blockchain systems focused solely on supply chain traceability, *FairChain* formalizes ABS compliance

and embeds stakeholder accountability through transparent smart contracts. The motivation behind choosing blockchain stems from its ability to provide decentralized control, cryptographic integrity, and auditability, all of which are critical in environments where multi-party trust, regulatory oversight, and data provenance are essential. By offloading sensitive data to encrypted off-chain storage and retaining only verifiable hashes on-chain, *FairChain* also mitigates privacy risks while preserving the immutability and traceability required for legal and regulatory purposes.

The remainder of this paper is structured as follows: Section 2 provides a background on the DDLC, Nagoya Protocol, and blockchain technology; Section 3 presents related work on model design; Section 4 describes the methodology, detailed architecture, and functions of our proposed model; finally, Section 5 provides a detailed discussion and concludes the article.

2. Background

This section provides a background on several of the research topics, including a background on the different stages of the DDLC, the UNDP NP, and the use of blockchain technology.

2.1. DDLC

Drugs usually undergo six developmental stages to guarantee their safety and efficiency and ensure that they will pass different regulatory requirements [12].

1. **Sample Collection:** Bio-samples are usually collected in areas far away from the laboratory; in order to make the most of these supplies and use them correctly, they must be properly preserved [4].
2. **Discovery and Development:** This phase usually consists of sample purification and screening [4]. In sample purification, cells from macroscopic samples or proteins are isolated using different purification protocols that usually require expertise and specific facilities [4]. During sample screening, the resulting compounds from sample purification are matched to potential targets in order to identify their activity [4]. Additionally, this process might be carried out using computational approaches [4].
3. **Preclinical Research:** Before a drug can be tested on humans, researchers must ensure that it will not cause serious harm to people [13]. To do so, drugs must first be tested on animals. Preclinical research must answer basic questions regarding safety, dosing, and toxicity levels [13].
4. **Clinical Research:** After the regulator (for instance, the Food and Drug Authority) reviews and approves preclinical research results, drugs are first tested on humans to ensure their safety and effectiveness [14]. Before clinical research begins, researchers must decide who qualifies to participate, how many people will participate, the duration of the research, how to limit research bias, how the drug will be provided to patients and at what dosage, what data will be collected, and how these data will be reviewed and analyzed [14]. Clinical research consists of the following four sequential phases, which are also explained briefly in Table 1:
    - In phase one, 20 to 100 volunteers who are healthy or have the condition/disease under study are tested for several months in order to answer questions regarding safety and dosage and to determine how much of the drug the human body can tolerate along with any acute side effects [14].
    - In phase two, several hundred people with the disease/condition are tested for several months to up to 2 years in order to find out more about the drug's efficacy and side effects [14].

- In phase three, 300 to 3000 volunteers who have the disease or condition are tested for 1 to 4 years in order to answer questions regarding long-term or rare side effects [14].
- In phase four, several thousand volunteers who have the disease/condition are tested to find out more about the drug's safety and efficacy [14].

5. **Regulator Review:** A drug developer must submit a new drug application to the regulator that includes all information about the drug, including preclinical and clinical research reports, proposed labeling, safety updates, drug abuse information, patent information, and directions for use [15]. After the regulator receives the new drug application request, it has 6 to 10 months to decide whether to approve the drug or deny it [15].
6. **Regulator Post-Market Drug Safety Monitoring:** Although clinical research results provide valuable and important information regarding drug safety and effectiveness, this information may change or be updated throughout the months and years that make up a drug's lifetime in the marketplace. After drugs become available for public use, the regulator continues to review and monitoring drug safety reports, and might agree to apply appropriate changes to dose or usage information or any other aspects [16].

Table 1. The four phases of clinical drug trials .

Phase No.	Number of Participants	State of Participants	Length of Study	Purpose	Outcomes
Phase 1	20–100	Healthy or people with the disease/condition	Several months	Safety and dosage	• How it works in the body • Acute side effects • Approximately 70% of drugs move to the next phase
Phase 2	Several hundred	People with the disease/condition	Several months to 2 years	Efficacy and side effects	Approximately 33% of drugs move to the next phase.
Phase 3	300 to 3000	People with the disease/condition	1 to 4 years	Efficacy and monitoring of adverse reactions	• Long-term or rare side effects • Approximately 25–30% of drugs move to the next phase
Phase 4	Several thousand	People with the disease/condition		Safety and efficacy	Regulator approval or denial

2.2. UNDP Nagoya Protocol for Access and Benefit-Sharing

According to [9], the Nagoya Protocol is defined as *"A transparent legal framework for the effective implementation of one of the three objectives of the Convention on Biological Diversity on access to genetic resources and the fair and equitable sharing of emerging benefits."* The Nagoya Protocol was adopted in Nagoya, Japan in 2010 [9], and aims to set up an institutional and legally binding mechanism to facilitate consistent and efficient adoption of the Access and Benefit-Sharing protocol at the global, national, and local levels [17]. Access and Benefit-Sharing refers to the general framework established by the Convention on Biological Diversity to promote fair and equitable sharing of benefits arising from the use of genetic resources. The Nagoya Protocol, adopted in 2010, is the specific legal instrument that provides a transparent and enforceable mechanism to implement Access and Benefit-Sharing at the international level. While Access and Benefit-Sharing is the broader principle, the Nagoya Protocol is the operational and legally binding protocol that enforces it.

By making sure that benefits of drug development are spread fairly, the Nagoya Protocol provides opportunities to sustainably preserve and use genetic capital, thereby strengthening the contributions of biodiversity to human growth and well-being [9].

2.3. Blockchain Technology

A blockchain can be defined as *"A time-stamped series of immutable records of data that is managed by a cluster of computers not owned by any single entity"* [18]. The data blocks that create a blockchain are secured and connected using cryptographic principles to form a single chain that is immutable, has no central authority (i.e., decentralized), and is shared and open for everyone with every party responsible for their own actions. In this way, blockchain systems can assure transparency [18]. A blockchain is a cost-effective means of transaction which makes it a simple yet ingenious automated way to safely transfer information from one endpoint to another [18]. A party in the transaction initiates the block creation process, and verification of this block is made via many computers (potentially up to millions). Finally, the chain is stored across the network, with uniqueness at both the record and history levels [18]. A blockchain works by the principle of duplication multiple times over the computer network where it is updated [18]. Because the chain is hosted by multiple computers, there is no single database location for storage [18]. Information held in a blockchain is generally and simultaneously accessible; hence, there is no centralized database to be hacked [18]. Current interest in blockchain technology mainly stems from its attributes of decentralization, immutability, transparency, and cryptographic data storage [18].

3. Related Work and Real-World Applications

The realm of research into environmental tracking solutions is diverse, as evidenced by the literature. One notable system is ItemTracker [19], which facilitates researchers in monitoring lab samples. ItemTracker caters to both small-scale and extensive operations through a flexible design that allows for the addition of diverse sample types. In addition, it provides comprehensive search capabilities for samples and their sources. Matrix Tracker [20] focuses on reducing errors in laboratory sample tracking, making it a vital tool in lab management. It stands out due to its configurability, which can be tailored to specific lab needs using its robust graphical configuration tools. Brooks Life Science [21], a leading name in this domain, offers innovative solutions in automated sample management, especially for drug discovery and storage applications. Their forte lies in providing top-notch solutions for automated compound and biological storage. Fluics Connect [22] stands as a champion for sustainable sample tracking in research labs, offering a slew of features such as smart labels and a robust mechanism to trace sample history. Additionally, recent work on secure privacy-preserving frameworks such as federated learning highlights the growing importance of protecting model integrity in healthcare and pharmaceutical settings, which aligns with FairChain's emphasis on traceable and tamper-resistant data sharing [23].

In addition to these commercial solutions, several academic efforts have explored blockchain's role in healthcare and pharmaceutical systems. Mettler [24] introduced early applications of blockchain in healthcare, emphasizing the value of decentralization and system trust. Yue et al. [25] proposed a blockchain-based healthcare data gateway that enhances privacy through encrypted data exchange. Benchoufi and Ravaud [26] demonstrated blockchain's role in improving clinical trial transparency and reproducibility. Azaria et al. [27] developed MedRec, a blockchain framework for managing electronic medical records that preserves patient ownership and access control. More recently, Hylock and Zeng [28] provided a comprehensive evaluation of blockchain systems in healthcare, advocating for alignment with legal and ethical standards.

Nevertheless, a common limitation among these solutions is their inability to provide an immutable record of sample originality along with their lack of comprehensive transparency for all stakeholders. While current solutions excel at tracking bio-samples and

presenting real-time status updates, they often fall short of the required level of trust among users. This highlights a gap in the literature, where there is a need for reliable bio-sample tracking solutions that can ensure transparency, efficacy, and user independence while minimizing human error and environmental impact. Furthermore, there is a pressing need to address the loss of originality and unfair distribution of profits, which demand more distinctive and equitable solutions.

FairChain bridges this gap by managing bio-sample history and updates while also ensuring transparency for all users. Unlike many of its counterparts, *FairChain* emphasizes a decentralized system without bias from pharmaceutical entities. It ensures the traceability of originality, is free of charge, verifies its users, and prioritizes the immutability of information. In sum, while several existing applications possess some features akin to our system's standards, there remains a distinct gap; the research domain urgently needs a bio-sample tracking system that prioritizes transparency, efficiency, and user autonomy while having minimal susceptibility to errors or environmental impacts. Addressing existing issues such as loss of originality and inequitable profit distribution enhances the value and distinctiveness of such a system.

Blockchain-Based Pharmaceutical Tracking Solutions

Several blockchain-based pharmaceutical tracking systems have been developed to enhance transparency, prevent counterfeit drugs, and ensure regulatory compliance. These solutions include MediLedger, IBM Pharma Ledger, VeChain ToolChain, PharmaLedger, BlockPharma, and Hyperledger Fabric. While these platforms have improved supply chain tracking, they primarily focus on pharmaceutical logistics rather than the entire DDLC.

- MediLedger enables secure data exchange among pharmaceutical supply chain participants, ensuring compliance with the Drug Supply Chain Security Act (DSCSA) in the U.S. [29].
- IBM Pharma Ledger provides end-to-end supply chain tracking with enterprise-grade blockchain encryption and data security [30].
- VeChain ToolChain™ integrates IoT-based tracking mechanisms (NFC and RFID) for real-time verification and anti-counterfeiting [31,32].
- BlockPharma offers QR code-based verification for patients to validate the authenticity of medications before purchase [33].
- Hyperledger Fabric, backed by The Linux Foundation, supports permissioned blockchain networks for enterprise pharmaceutical applications, and has been adopted by companies such as Pfizer and Merck [34].

While these solutions focus on logistics, *FairChain* uniquely integrates blockchain into the entire DDLC, ensuring not only supply chain security but also compliance with the Nagoya Protocol. This enables fair benefit sharing, transparent drug development, and bio-sample traceability, features that are absent in existing solutions. A comparison of blockchain-based pharmaceutical tracking solutions is summarized in Table 2.

Table 2 presents a comparative analysis of blockchain-based pharmaceutical tracking solutions. While MediLedger, IBM Pharma Ledger, VeChain, PharmaLedger, BlockPharma, and Hyperledger Fabric enhance drug security, transparency, and regulatory compliance, *FairChain* uniquely extends beyond supply chain tracking to cover the entire DDLC, ensuring fair benefit-sharing through compliance with the Nagoya Protocol. Moreover, *FairChain* uniquely embeds the compliance obligations of the Nagoya Protocol within its system design, addressing legal interoperability and benefit-sharing traceability.

Table 2. Comparison of blockchain-based pharmaceutical tracking solutions.

Feature	Medi-Ledger	IBM Pharma Ledger	VeChain ToolChain	Pharma-Ledger	Block-Pharma	Hyper-Ledger Fabric	*FairChain*
Primary Focus	Supply chain tracking	Supply chain and compliance	Supply chain and IoT	Clinical trials and patient consent	Drug authentication	Enterprise pharma blockchain	Full DDLC and Nagoya Protocol compliance.
Security	Permissioned blockchain	Enterprise encryption	IoT tracking (NFC and RFID)	Secure patient data sharing	QR-code verification	Permissioned enterprise blockchain	Public and permissioned blockchain, smart contracts
Scalability	DSCSA-compliant transactions	Enterprise scalability	IoT-enabled real-time tracking	GDPR and pharma compliance	Consumer mobile integration	High-performance ledger transactions	High-volume transactions with decentralized governance.
Regulatory Compliance	DSCSA (U.S.)	Global pharma regulations	Pharma supply chain laws	GDPR and EU pharma laws	Anti-counterfeit regulations	Pharma industry compliance	Nagoya Protocol for bio-sample tracking.
Stakeholders	Manufacturers, wholesalers, regulators	Pharma companies, regulators	Supply chain logistics, pharma firms	Researchers, patients, regulators	Pharmacies, patients	Pharma companies, regulators	Researchers, landowners, pharma firms, regulators.
Consensus Mechanism	Permissioned (custom)	PoS (IBM)	PoA	Not specified	Not specified	Raft/PBFT	PoS (Ethereum)
Data Privacy Model	Encrypted messaging	Private enterprise data	Encrypted IoT streams	GDPR consent tools	QR-code validation	Encrypted permissioned ledger	Off-chain storage with hashed pointers

4. Methodology

The development of *FairChain* followed a user-centered design approach supported by expert consultations, a literature review, and iterative modeling. The goal was to ensure that both the technical architecture and policy-aligned mechanisms would satisfy both compliance with the Nagoya Protocol and the practical needs of the Saudi pharmaceutical research environment.

4.1. Stakeholder Engagement and Expert Interviews

To ensure relevance and regulatory alignment, we conducted two rounds of semi-structured interviews with senior experts: Prof. Sara Al-Rashood and Mr. Abdulmohsen Abanumai. Prof. Al-Rashood is a Professor of Pharmaceutical Chemistry at the College of Pharmacy, King Saud University, Saudi Arabia, and a Drug Design Consultant at the Saudi Food and Drug Authority (SFDA). Mr. Abanumai is a Strategic Solutions Senior Manager at the National Unified Procurement Company (NUPCO), Saudi Arabia, and an expert in clinical pharmacy. Both experts have over 15 years of experience in pharmaceutical research and legal compliance. Participants were selected through purposive sampling due to their direct involvement in national regulatory planning and Access and Benefit-Sharing (ABS) governance.

The interviews aimed to:

- Validate the compliance bottlenecks faced by researchers and institutions.
- Identify critical trust, transparency, and auditability features expected in a blockchain ecosystem.
- Understand institutional concerns related to data governance, IP attribution, and ABS traceability.

Each session lasted approximately 60 min and followed a consistent questionnaire format. Notes were transcribed and coded using thematic analysis to extract actionable system requirements.

4.2. System Requirements and Development Inputs

Three data sources guided the functional specification:

1. Interviews with subject matter experts.
2. A document review of national ABS frameworks, Nagoya Protocol requirements, and guidance from ABS Clearing-House.
3. Technical architecture benchmarking of similar blockchain systems in healthcare and digital identity governance.

The stakeholder feedback informed smart contract role design, metadata structures, and ledger logging requirements.

4.3. Bias and Scope Limitations

We acknowledge that the small sample size (n = 2 experts) may limit generalizability; however, both experts held national regulatory roles, offering deep insights into the practical barriers of Nagoya Protocol implementation in research settings.

Additionally, the current system represents a prototype. While design elements are complete, further pilot testing and simulation are needed to evaluate system performance, usability, and scalability. The resulting system-level requirements were synthesized into a summary of design contributions, which are detailed in Table 3.

Table 3. Summary of *FairChain*'s key technical contributions.

Feature	*FairChain* Contribution
Trust Framework	Verifiable smart contracts with multi-role agents
Auditability	Immutable on-chain provenance using blockchain ledger
Legal Interoperability	Smart contract metadata aligned with Nagoya Protocol terms
Transparency	Executable trace logs and tokenized resource tracking
Modularity	Composable architecture supporting integration with research and IP workflows

5. *FairChain* Functional Design

The design of *FairChain* consists of the following key components: system users, architecture, database, and user interface designs. Each component is explained further below.

5.1. User Roles

FairChain is designed for the following user categories: pharmaceutical company representatives (*Pharmaceutical* for short), reviewers (*Reviewer* for short), bio-sample landowners (*Landowner* for short), and healthcare practitioners (*Doctor* for short). All system users are able to login using their usernames and passwords for authentication and authorization (with their main credentials issued by an administrator), while *Landowner* users login using

only their national identification credentials. The *Pharmaceutical* category represents the main users of *FairChain*, who are responsible for registering new bio-samples and assigning them to a *Landowner* as well as for registering a new drug, recording preclinical research and labeling data, and finally applying for a New Drug Application request. The *Pharmaceutical* user is a skilled person with high technical expertise. Both *Doctor* and *Reviewer* user categories also consist of highly skilled users with technical expertise. *Reviewer* users are responsible for approving or denying a New Drug Application request and viewing the bio-sample ledger, while *Doctor* users mainly record clinical research data. Users in the *Landowner* category use *FairChain* to view the bio-sample ledger, and are expected to have low- to-mid-level technical skills. The full use case showing all the actors and respective cases is illustrated in Figure 2.

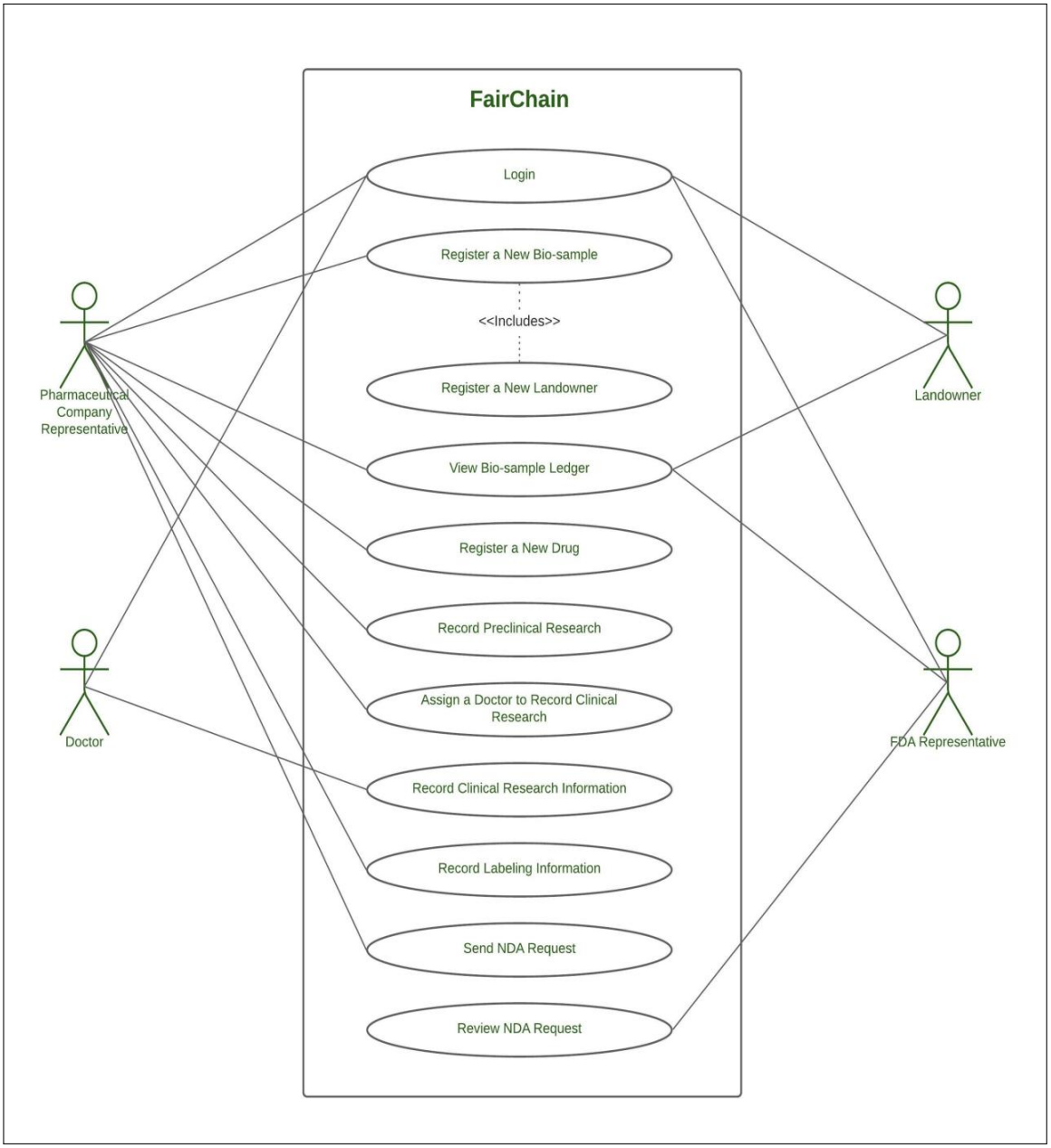


Figure 2. *FairChain* use case diagram.

5.2. Functional Requirements

The *FairChain* system supports the following user-driven functional actions:

- **Login:** Users (including pharmaceutical company representatives, FDA representatives (reviewers), doctors, and landowners) can log into the system. The system prompts them to enter credentials, which are then validated. If correct, the user gains access; otherwise, the login is denied.
- **Register a New Bio-Sample:** A pharmaceutical company representative records a new bio-sample by entering details such as its name, type, and geographical location. The system verifies and stores the information, updating the bio-sample ledger on the blockchain. The sample is also assigned to a landowner.
- **Register a New Landowner:** A pharmaceutical company representative registers a landowner by entering identification details, contact information, and address. The system validates and stores this information, linking the landowner to a previously registered bio-sample.
- **View Bio-Sample Ledger (Pharmaceutical Company/FDA Representative):** Pharmaceutical company and FDA representatives can view the bio-sample ledger, tracking all changes from registration to the latest updates. The system retrieves and displays this timeline, ensuring transparency.
- **View Bio-Sample Ledger (Landowner):** Landowners can log into the system and view their assigned bio-sample ledger, tracking all updates from registration onward. The system ensures that access is restricted to authorized users.
- **Register a New Drug:** A pharmaceutical company representative registers a new drug by providing its name and active ingredients. The system verifies and stores this information, updating the bio-sample ledger.
- **Record Preclinical Research:** A pharmaceutical company representative records preclinical research data, including the type and number of test subjects, toxicity levels, and human testing eligibility. The system validates and saves the data, updating the blockchain ledger.
- **Assign a Doctor to Record Clinical Research:** A pharmaceutical company representative assigns a doctor to clinical research by entering the doctor's ID. After confirmation, the system grants access to record clinical trial data.
- **Record Clinical Research:** An authorized doctor enters clinical research data, including trial phases, participant details, study length, and outcomes. The system verifies and updates the bio-sample ledger accordingly.
- **Record Labeling Information:** A pharmaceutical company representative enters drug labeling details such as indications, usage, dosage, and adverse reactions. The system validates and stores the information, updating the blockchain.
- **Send New Drug Application (NDA) Request:** A pharmaceutical company representative submits an NDA to the FDA, including safety updates, drug abuse information, and patent details. The system imports necessary data, verifies inputs, and updates the NDA status.
- **Review NDA:** An FDA representative reviews a submitted NDA request and determines whether to approve or deny it. The system updates the NDA status accordingly, ensuring transparency in the approval process.

5.3. Non-Functional Requirements (NFRs)

The following Non-Functional Requirements (NFRs) were defined as target design goals to guide system development. While implementation is ongoing, these goals reflect the intended performance, security, and legal compliance properties:

- **Performance:** The platform is designed to support efficient contract execution and sub-second response times for read operations under typical load conditions.

- **Scalability:** *FairChain* is architectured for horizontal scalability across research institutions by using modular agents and smart contracts.
- **Security:** Role-based access control and cryptographic signing are built into the contract deployment and token issuance mechanisms.
- **Auditability:** Each transaction is immutably logged to ensure transparency and verifiability during post hoc compliance reviews.
- **Legal Interoperability:** Metadata schemas are aligned with Nagoya Protocol terminologies to support traceable benefit-sharing reporting.

Future deployments will validate these NFRs under simulated regulatory and scientific workflows. Figure 3 illustrates how the requirements (functional and NFRs) support *FairChain*.

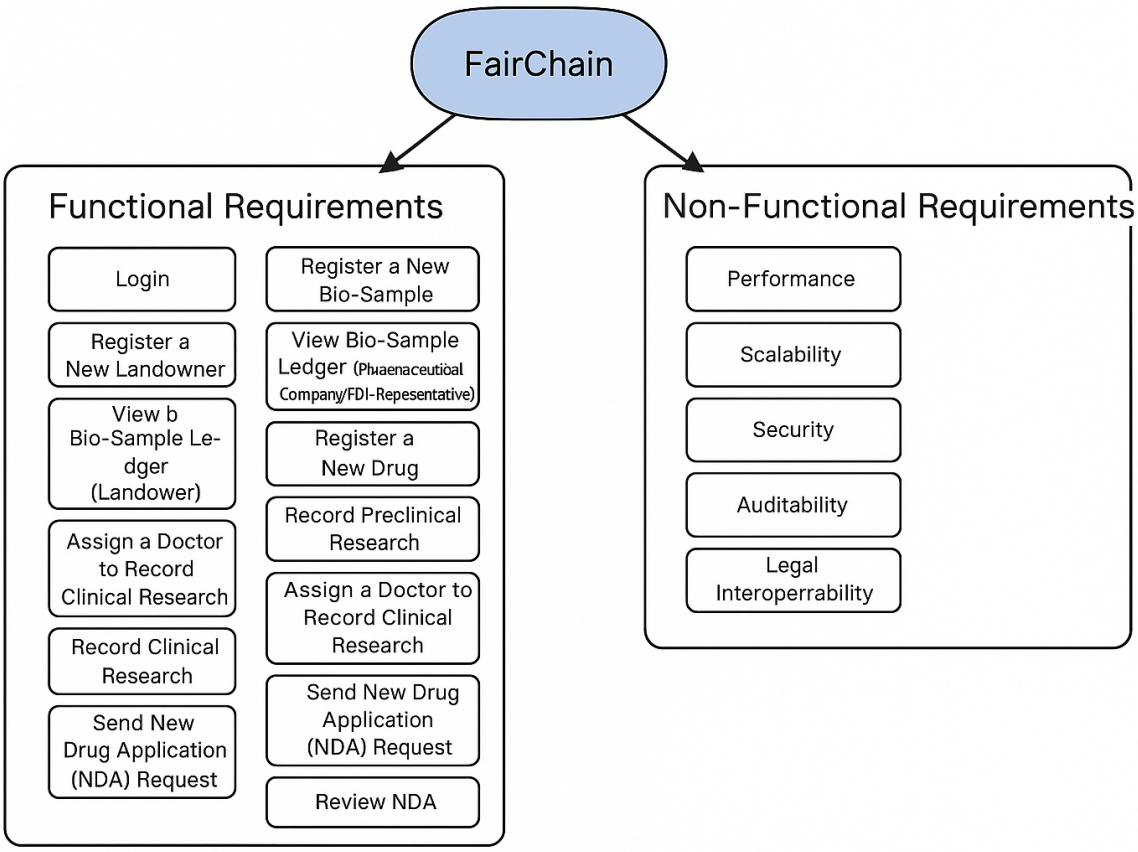


Figure 3. Mapping functional and non-functional requirements: diagram illustrating how each requirement supports the *FairChain* system.

6. *FairChain* Design

This section presents the technical design of the *FairChain* platform, detailing how the system architecture, database, user interfaces, and workflows were derived from stakeholder needs and methodological insights. The design emphasizes scalability, security, and usability for key stakeholders across the DDLC. The following subsections describe the modular components and interaction flow that work together to enable *FairChain*'s functionality, from secure data handling to interface usability.

6.1. FairChain Architecture Design

Given the users and features of *FairChain*, the most suitable design approach consists a layered architecture providing high-level information security capabilities and enhanced scalability. The design of *FairChain*'s architecture consists of the following layers (illustrated in Figure 4):

- **User Interface Layer**: This layer serves as the point of interaction between users and the system, providing an accessible and user-friendly interface. It encompasses elements such as dashboards, forms, and other visual components that facilitate user engagement with the underlying application functionalities.
- **Application Layer**: Also known as the service layer, this component contains the core business logic and rules of the system. It processes user inputs received from the User Interface Layer, executes specific operations, and manages data flows between the user interface and the data storage components.
- **Authentication Layer**: This security-focused layer is responsible for verifying user identities and controlling access to system resources. It ensures that only authorized users can access certain functionalities or data, typically through mechanisms such as login credentials, tokens, or multi-factor authentication.
- **Blockchain Layer**: Serving as the decentralized ledger, this layer records all transactions and data entries in a secure and immutable manner. It operates on the Ethereum blockchain using the Proof-of-Stake (PoS) consensus mechanism, which enhances energy efficiency and network security while supporting scalability. This layer ensures transparency and trust by maintaining a tamper-proof history of activities validated by staked validators.
- **Database Layer**: This layer is responsible for the structured storage, retrieval, and management of data within the system. It handles operations such as querying, updating, and organizing data to support the application's requirements, thereby ensuring efficient data access and integrity.

These layers work collaboratively to create a robust, secure, and user-friendly system architecture.

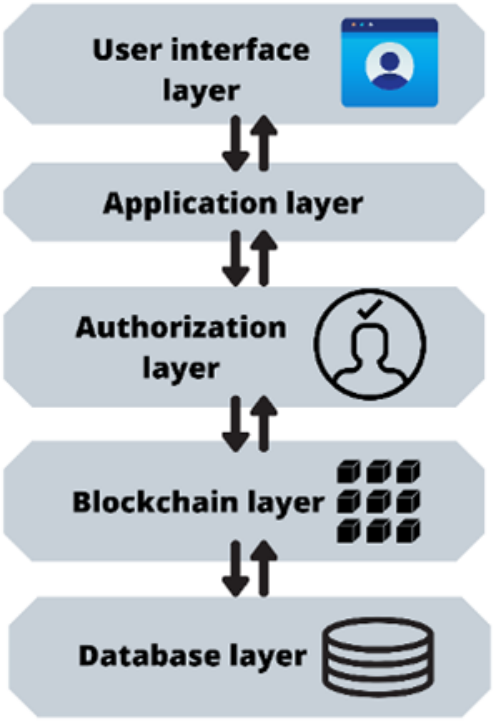


Figure 4. *FairChain* architecture: a modular multi-layered system design including the User Interface Layer, Application Layer, Authentication Layer, Blockchain Layer, and Database Layer. These layers work in tandem to ensure scalability, trust, and security across all DDLC stages.

6.2. Database Design

The database design for *FairChain* is structured to efficiently manage and organize data related to landowners, bio-samples, and drugs. The Entity–Relationship (ER) core entities, their attributes, and the relationships among them ensure data integrity and optimized retrieval (see Figure 5). The ER diagram represents a pharmaceutical drug development ecosystem involving multiple user roles such as FDA Representatives, Pharmaceutical Company Representatives, Doctors, and Landowners, each inheriting from a general User entity. The workflow illustrates how BioSamples are owned and registered, leading to drug creation, followed by clinical and preclinical research phases, NDA submissions, and drug labeling. Key entities like Drug, ClinicalResearchPhase, and NDA have detailed attributes (e.g., side effects, dosage form, toxicity level) and are interconnected through defined relationships. Several one-to-many (1:*) relationships are present—for instance, a single Doctor can conduct multiple ClinicalResearchPhases, and one Drug can be associated with multiple Labeling records, reflecting variations in dosage forms and usage indications.

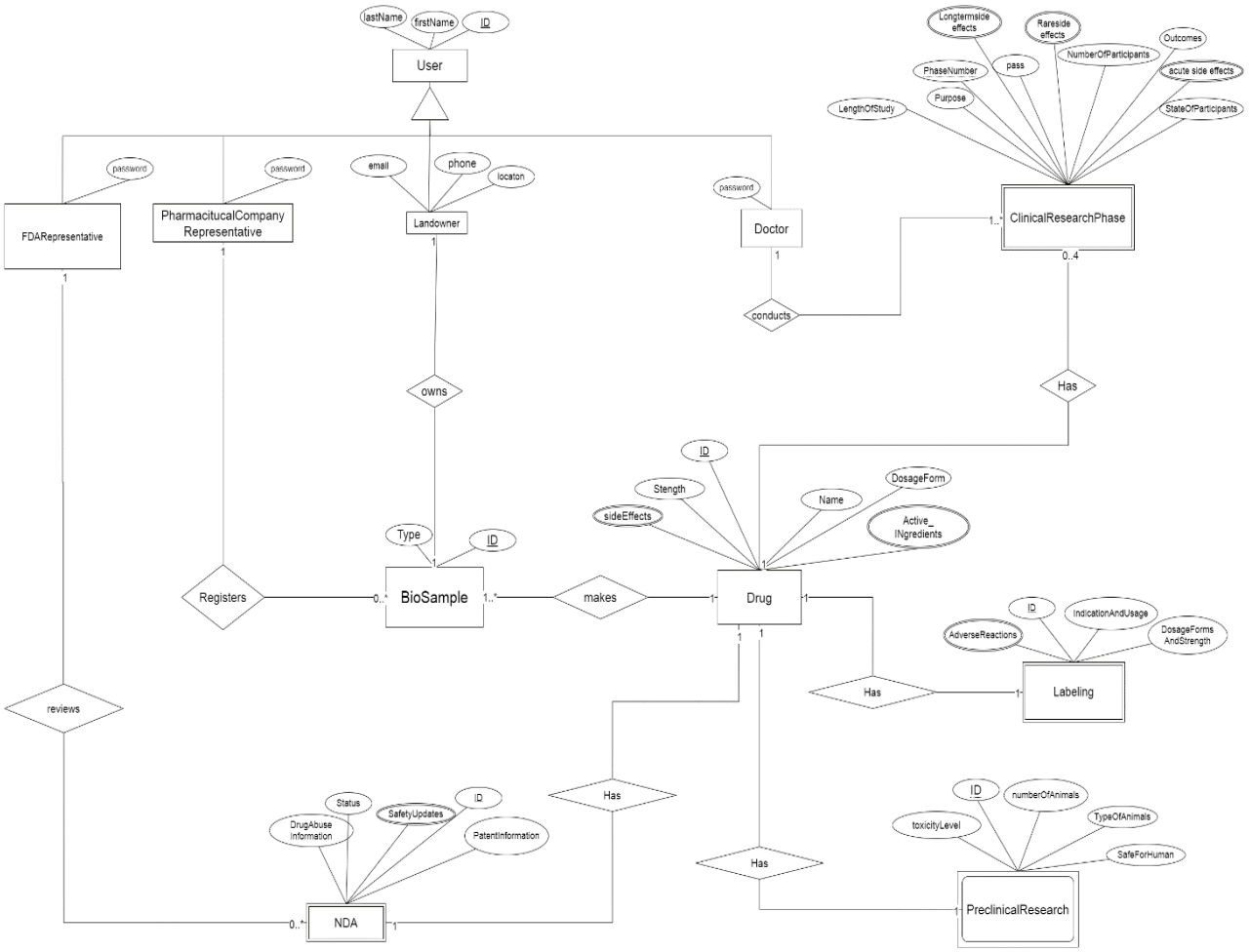


Figure 5. *FairChain* ER diagram: representing a pharmaceutical drug development ecosystem conceptual model involving core entities, their attributes, and the relationships among them to ensure data integrity and optimized retrieval.

The schema design (Figure 6) translates this conceptual model into a relational database structure, specifying table definitions, primary and foreign key constraints, and indexing strategies to enhance query performance. Each entity is designed with normalization principles to minimize redundancy and maintain data consistency. The landowner entity captures ownership details, the bio-sample entity records sample-related metadata,

and the drug entity manages pharmaceutical information. The relationships among these entities are structured using foreign key constraints to enforce referential integrity, ensuring reliable data linkage across the application. This design supports efficient data operations while ensuring scalability and maintainability for future enhancements.

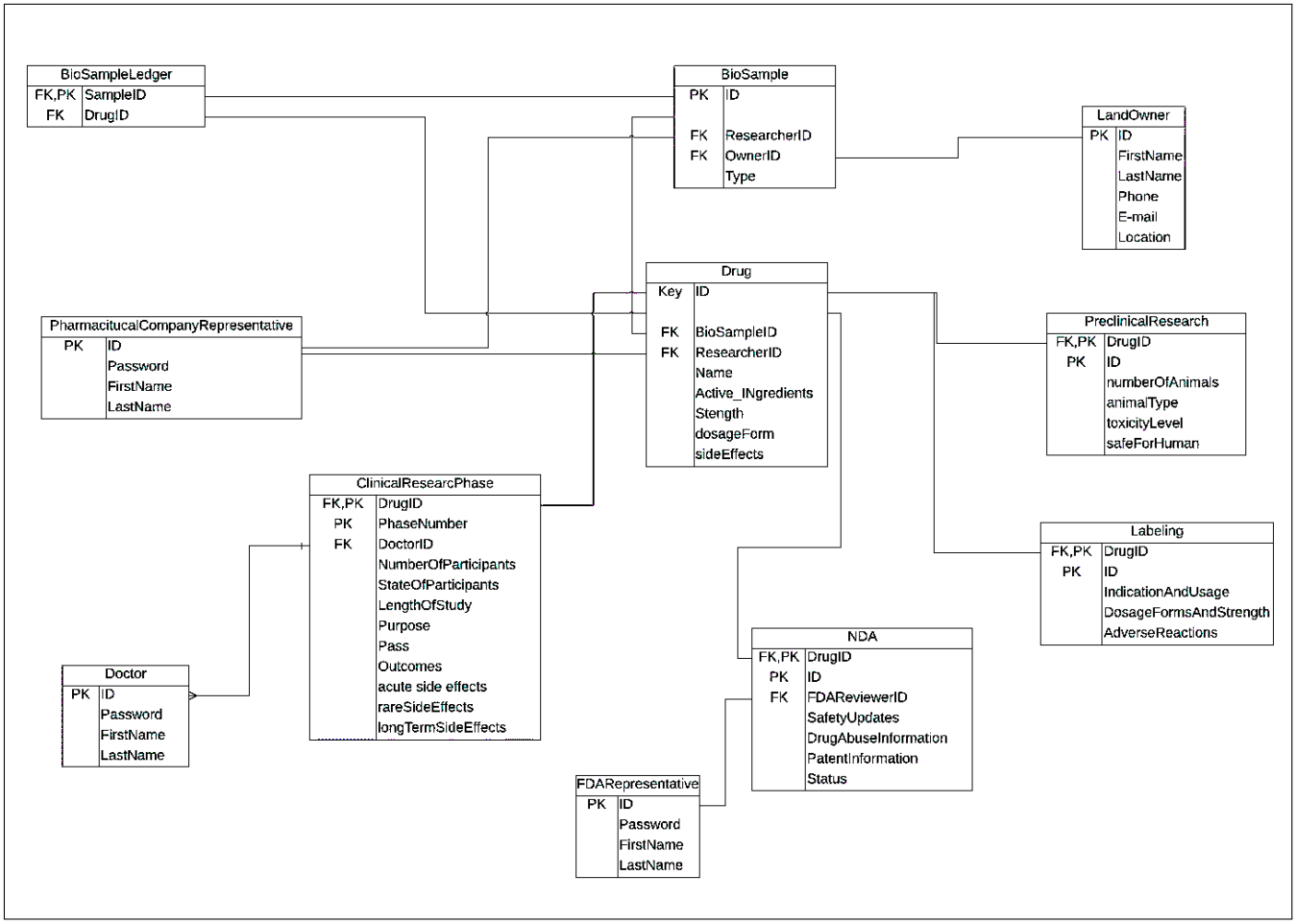


Figure 6. *FairChain* database schema: relational structure showing core entities such as Landowner, Bio-Sample, and Drug along with their key attributes and foreign key relationships. The database schema is designed for efficient and scalable storage of off-chain data while supporting auditability.

6.3. FairChain System Screen Flow

The system's screen flow design is structured to facilitate seamless navigation and efficient task execution across different user roles. The User Interface Design ensures clarity and accessibility, starting with the Login screen, which serves as the entry point for all user dashboards. Moreover, each user role (Pharmaceutical Company Representative, FDA Representative, Landowner, Doctor) is provided with a distinct interface tailored to their respective functionalities. The logical structuring of functionalities across screens allows for streamlined interactions, reducing errors and improving the overall efficiency of pharmaceutical research and approval workflows (See Figure 7).

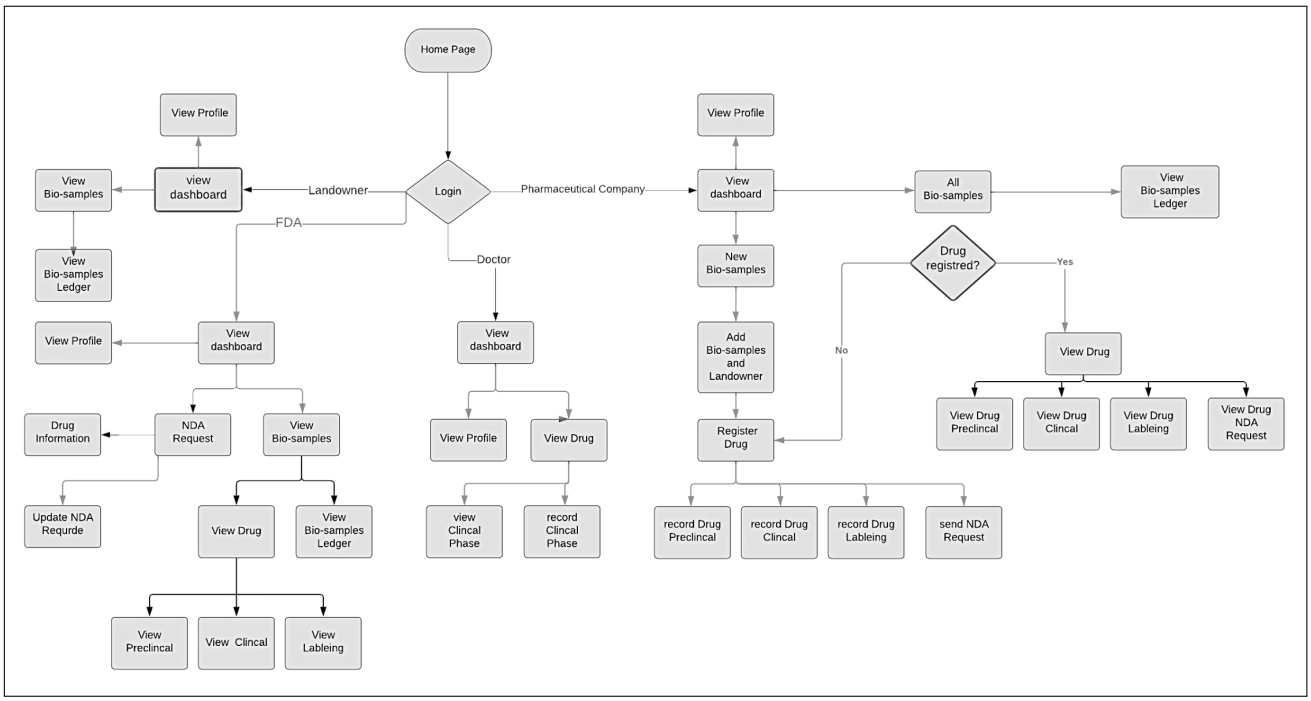


Figure 7. *FairChain* user screen flow diagram depicting the navigation logic and screen transitions based on user roles (Pharmaceutical Company, Reviewer, Doctor, Landowner) to support the decentralized recording of drug development activities.

Example Workflow Scenario: A Pharmaceutical Company Representative begins by registering a new plant-derived bio-sample and assigning a Landowner. Preclinical research is recorded and linked to the bio-sample. A Doctor enters clinical research data, followed by a New Drug Application (NDA) submission. An FDA Reviewer then evaluates the request and updates its approval status, while the Landowner is able to transparently monitor the entire chain of events through the ledger interface.

6.4. User Interface Design

Thw *FairChain* application includes the design of a dedicated user interface for each of the user categories described in Section 5.1.

The *FairChain* user interface includes several core functions tailored to different stakeholders, ensuring efficient workflow management and data transparency. For pharmaceutical representatives, the system enables the submission of NDA requests (Figure 8), allowing them to initiate the regulatory approval process. Regulators such as FDA representatives can efficiently view received NDA requests (Figure 9) and access all registered bio-samples (Figure 10), facilitating oversight and verification. For doctors, the interface provides a function to record clinical research phase information (Figure 11), ensuring proper documentation of trial progress. These key functions enhance usability, streamline operations, and support the secure and transparent management of bio-sample and pharmaceutical data within *FairChain*.

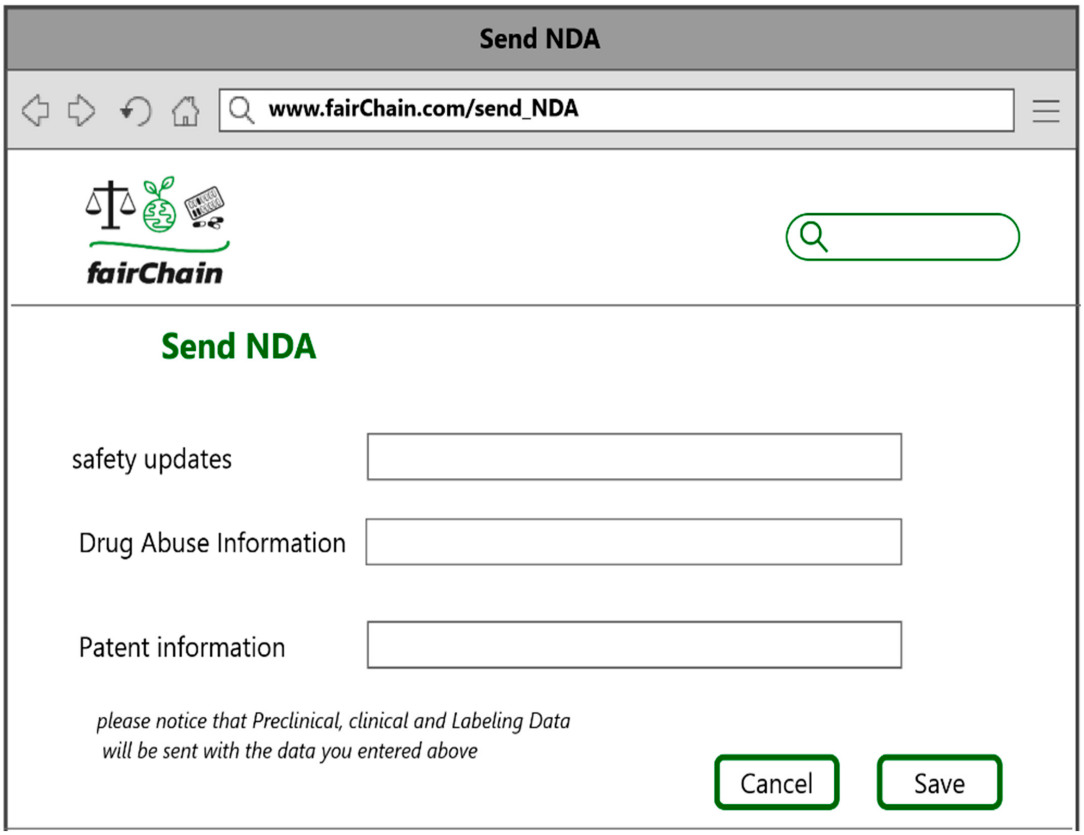


Figure 8. Send NDA Request page: interface view for Pharmaceutical Company Representatives to submit New Drug Applications (NDAs), including attached metadata and clinical data summaries for regulator review.

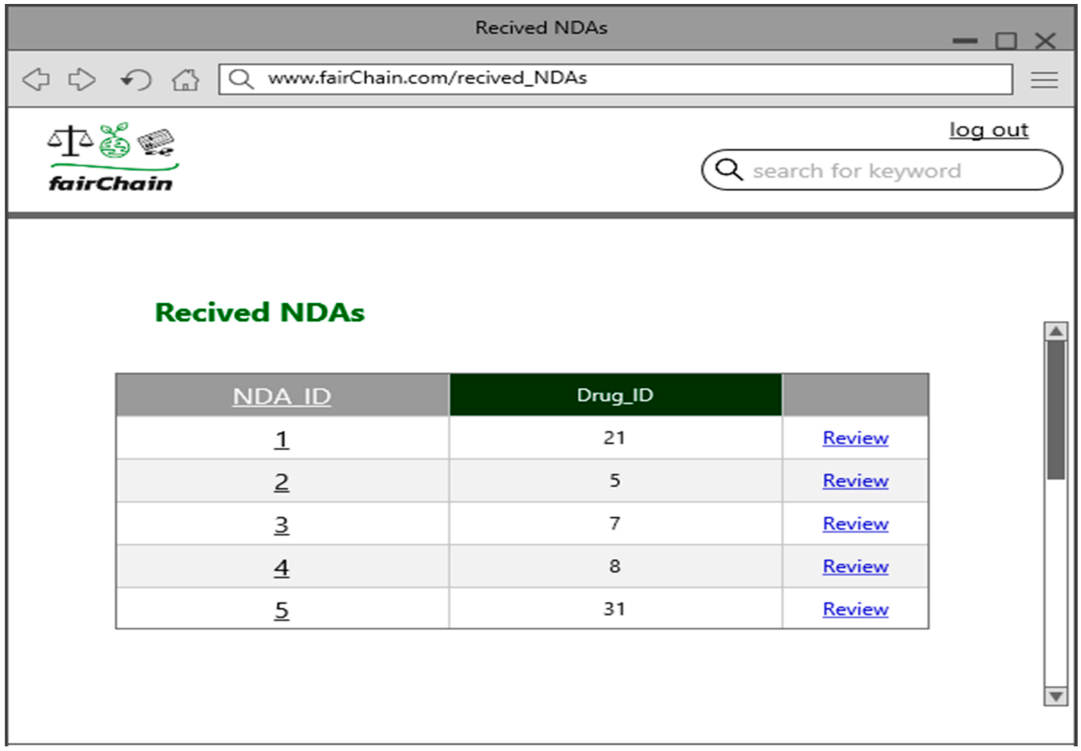


Figure 9. View Received NDA Requests page: interface used by Reviewers (e.g., FDA officials) to review and act upon pending NDAs submitted through the *FairChain* system.

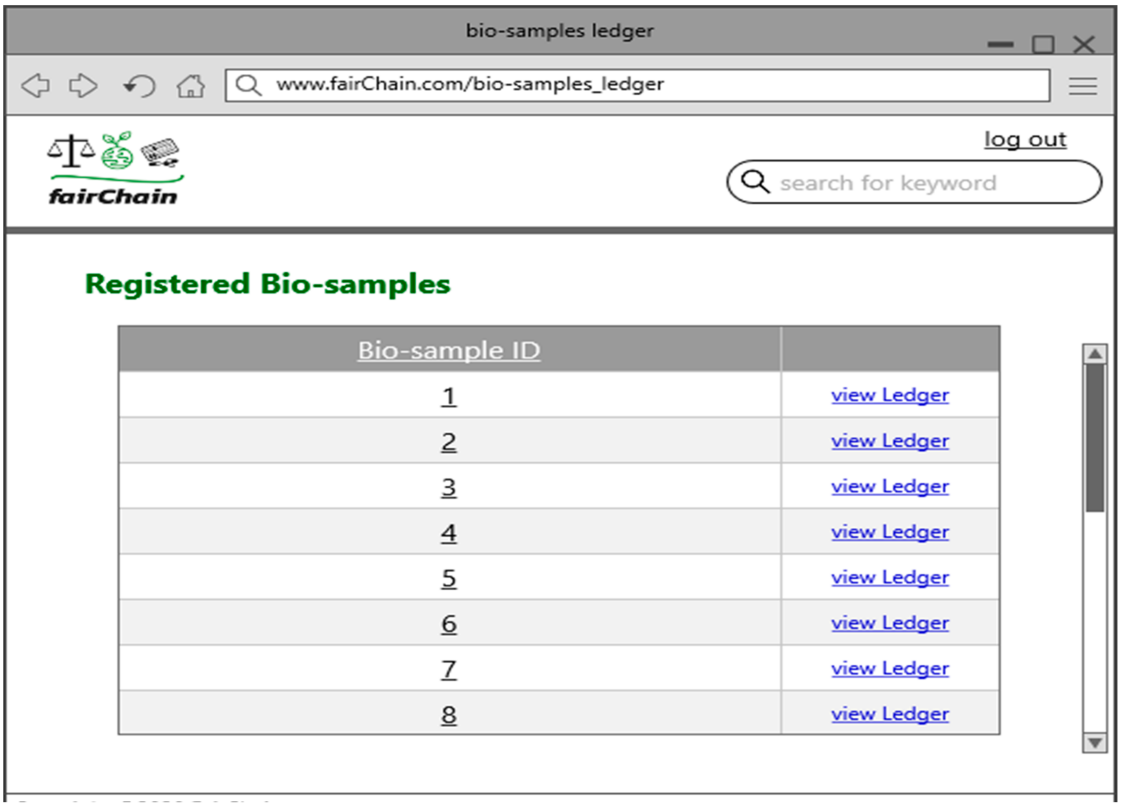


Figure 10. View All Registered Bio-Samples page: Regulators can view all bio-samples logged on-chain and off-chain, enabling auditability and traceability of source material throughout the DDLC.

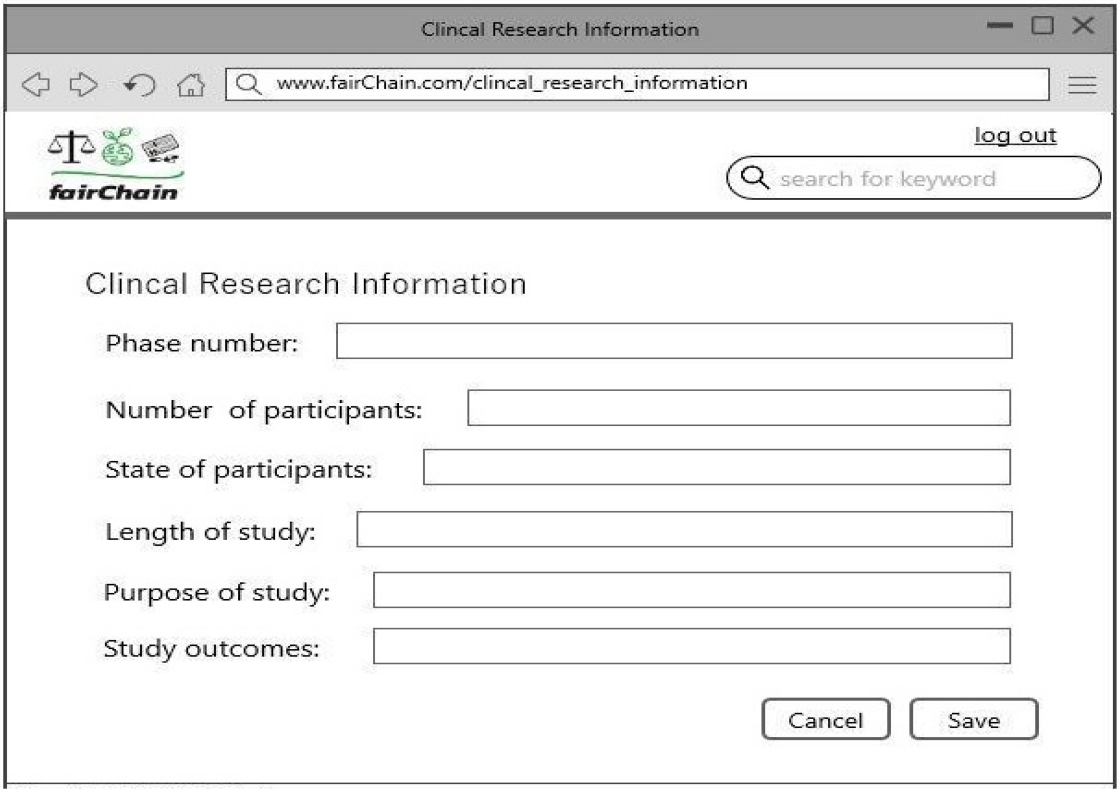


Figure 11. Record Clinical Research Phase Information page: Doctors can view this interface and record clinical trial data across different phases as part of the drug approval process. All entries are logged and timestamped for transparency.

7. *FairChain* Implementation

The *FairChain* implementation encompasses the integration of key components necessary for secure and efficient data management. It includes database implementation, which handles off-chain data using *phpMyAdmin*, and blockchain ledger integration, which leverages *Ethereum* smart contracts for transparency and security. Additionally, the web application development focuses on creating a user-friendly interface, while system integration ensures seamless communication between the database and blockchain using *Web3.js*. The following sections provide a detailed breakdown of each component.

7.1. Database Implementation

In this part, we have used *phpMyAdmin*, "a free software tool written in *PHP*, intended to handle the administration of *MySQL* over the Web". *phpMyAdmin* was used to manage the database and its columns, relations, indexes, users, permissions, etc. *phpMyAdmin* operations can be performed via the user interface while retaining the ability to directly execute any *SQL* statement. *phpMyAdmin* was used to store off-chain data [35]. The SQL scripts for creating the Landowner, Bio-sample, and Drug tables are presented in Figures 12–14, detailing the implementation of the database ER and schema presented in Figures 5 and 6.

```
48 CREATE TABLE `landowner` (
49   `id` int(11) NOT NULL,
50   `name` varchar(50) NOT NULL,
51   `username` varchar(50) NOT NULL,
52   `email` varchar(50) NOT NULL,
53   `password` varchar(50) NOT NULL,
54   `mobile` varchar(50) NOT NULL,
55   `id_num` int(11) NOT NULL,
56   `location` varchar(255) NOT NULL
57 )
```

Figure 12. *Landowner* table creation.

```
89 CREATE TABLE `sample` (
90   `id` int(11) NOT NULL,
91   `name` varchar(50) NOT NULL,
92   `type` varchar(50) NOT NULL,
93   `location` varchar(255) NOT NULL,
94   `landowner_id` int(11) NOT NULL
95 )
```

Figure 13. Bio-sample table creation.

```
31 CREATE TABLE `drug` (
32   `id` int(11) NOT NULL,
33   `name` varchar(50) NOT NULL,
34   `active_ingredients` varchar(255) NOT NULL,
35   `strength` varchar(50) NOT NULL,
36   `dosage_form` varchar(50) NOT NULL,
37   `side_effects` varchar(255) NOT NULL,
38   `sample_id` int(11) NOT NULL
39 )
```

Figure 14. Drug table creation.

Due to the high complexity and size of drug lifecycle datasets, *FairChain* avoids storing such data directly on-chain; instead, the system logs cryptographic hashes and metadata pointers to these datasets, which are securely maintained in an off-chain MySQL database. This hybrid approach ensures that the blockchain remains lightweight while preserving verifiability and auditability of lifecycle events.

To ensure the confidentiality of sensitive pharmaceutical and patient data, *FairChain* employs a hybrid data architecture that separates public blockchain activity from private data management. Personally identifiable information, clinical trial records, and proprietary drug data are never stored directly on the blockchain. Instead, such sensitive data are securely maintained in an off-chain relational database managed through *phpMyAdmin* and protected by encryption, authentication protocols, and role-based access controls. Only cryptographic hashes and essential metadata are recorded on-chain. This ensures integrity and traceability without disclosing confidential content.

This design aligns with data protection regulations such as the General Data Protection Regulation (GDPR) and Health Insurance Portability and Accountability Act (HIPPA) while supporting core principles such as data minimization and purpose limitation. Moreover, the off-chain structure allows for the enforcement of data subject rights including access, rectification, and erasure, thereby ensuring that *FairChain* maintains transparency and auditability while fully complying with privacy requirements.

7.2. Blockchain Ledger Implementation

To deploy blockchain technology into the *FairChain* system, a smart contract was developed using *Solidity*, the programming language provided by the *Ethereum* public blockchain. The smart contract is responsible for processing user transactions, ensuring data integrity, and enabling decentralized verification. Through this implementation, critical operations such as bio-sample registration, NDA updates, and block creation are securely managed on the blockchain.

The implementation process began with designing a smart contract that encapsulates the core functionalities of *FairChain* (see Figures 5 and 6). The One of the fundamental components of the contract is the mapping mechanism (Figure 15), which efficiently associates bio-sample records with their respective owners and metadata. The contract includes a function called Add BioSample, which allows authorized users to register new bio-samples on the blockchain (Figure 16). Additionally, the contract provides an Update NDA function, which enables modifications to the NDA status of a specific bio-sample while ensuring immutability and traceability (Figure 17).

FairChain is currently deployed on the Ethereum mainnet, which employs Ethereum's PoS consensus algorithm. Validators are selected based on the quantity of cryptocurrency they stake, which promotes honest behavior and discourages malicious activity through economic penalties. Only transactions validated by these staked participants are appended to the blockchain. This consensus model offers robust security while significantly reducing energy consumption compared to traditional Proof-of-Work (PoW) mechanisms, thereby supporting sustainability in blockchain deployment. The use of PoS aligns with *FairChain*'s goals of achieving both transparency and environmental responsibility. While this implementation targets public Ethereum, future iterations may explore permissioned blockchains with configurable consensus protocols such as Practical Byzantine Fault Tolerance (PBFT) (a consensus algorithm designed for distributed systems—especially permissioned blockchains—where all participants are known and partially trusted) for enterprise-level applications.

```
24          //mapping bio samples to their ids
25        mapping (int =>BioSample) BioSampless;
26
27      //similar to array of bio samples
28        int[] public BioSamplesIds;
29
30        //count bio samples
31        function countBioSamples() view public returns(uint){
32              return BioSamplesIds.length;
33            }
```

Figure 15. *FairChain* mapping.

```
38    function addBioSample( int _id, string memory _name ,string memory _location, int _landowner)  public {
39        BioSample storage bioSample= BioSampless[_id];
40        bioSample.id=_id;
41        bioSample.name=_name;
42        bioSample.location=_location;
43        bioSample.landowner=_landowner;
44        //other values can be empty
45        BioSamplesIds.push(_id);
46    }
```

Figure 16. Adding a new bio-sample.

```
111         function updateNda(int _id, int _ndastatus)  public{
112               BioSample storage bioSample=BioSampless[1];
113               if (bioSample.drugId==_id) return;
114             for (uint j=0 ; j<=countBioSamples();j++){
115                 bioSample= BioSampless[int(j)];
116                 if (bioSample.drugId==_id)
117                 break;
118             }
```

Figure 17. Updating an NDA request.

After a transaction is executed, the blockchain processes and validates the request, leading to the creation of a new block. This block encapsulates the transaction details, ensuring transparency and security through cryptographic hashing. The details of this block creation process are recorded and stored on the blockchain, including the transaction metadata and unique identifiers, thereby preserving the integrity of all system interactions.

To ensure transparency and traceability while maintaining privacy, *FairChain* records a selected set of metadata for each registered bio-sample on the blockchain. This includes a unique sample identifier, a cryptographic hash of its complete off-chain record, the timestamp of registration, the general region of origin, the sample category (e.g., plant or microbial), the landowner's reference ID, and a status indicator. These metadata support verifiability while preserving confidentiality, as all sensitive biological and personal data are maintained off-chain in a secure relational database.

By leveraging blockchain technology, *FairChain* ensures that bio-sample data remain tamper-proof, auditable, and verifiable, thereby enhancing trust between stakeholders. The decentralized nature of the Ethereum blockchain mitigates risks associated with data manipulation while providing a transparent and secure mechanism for managing sensitive pharmaceutical and research-related information.

To mitigate the impact of blockchain overhead such as latency, gas fees, and throughput limitations, *FairChain* is designed to record only essential transaction metadata on-chain. All computationally intensive or voluminous data (e.g., trial data or analytical results) are stored off-chain. Smart contracts were developed with gas efficiency in mind, and future iterations of *FairChain* will explore the integration of layer 2 scaling solutions that are built

on top of layer 1 base blockchain and enterprise-grade blockchains. This would help to enhance performance in general through improved scalability, reduced transaction costs (gas fees), and increased speed without compromising the security of the main blockchain. To mitigate risks associated with the centralized MySQL database, future iterations of *FairChain* will implement replication and failover clustering for high availability. Encrypted off-site backups will also be enabled. In addition, we are exploring the use of decentralized storage solutions such as IPFS for storing hashed data chunks or migrating metadata to permissioned blockchain environments such as Hyperledger Fabric.

The smart contracts implemented in *FairChain* are custom-built using Solidity and deployed on the Ethereum blockchain. They function as workflow automation and data registry contracts, enabling trusted operations such as bio-sample registration, ownership tracking, clinical milestone verification, and NDA review status updates. These contracts are designed to interact only with authorized user roles and to maintain an immutable ledger of key drug development events, leaving sensitive data securely managed in the off-chain database.

The smart contract was tested extensively on the Goerli Ethereum testnet, which provides a safe and cost-free environment for evaluating blockchain-based applications. Functional validation was carried out using Remix IDE, and the Truffle framework was employed for unit testing of key smart contract functions such as bio-sample registration and NDA updates. Additionally, the *Web3.js* interface was used to simulate full-stack interactions. All transactions were monitored on Goerli Etherscan to verify successful event emissions, block creation, and access control behavior. No deployment to the Ethereum mainnet has been performed at this stage, pending a full system evaluation and security auditing.

The correctness of the *FairChain* smart contract was validated through manual code inspection, unit testing, and functional integration tests using Remix IDE and the Truffle framework. These evaluations ensured that the access controls, transaction logic, and blockchain state changes operate as expected. Although formal verification tools were not employed in this prototype, future versions of *FairChain* will explore integrating static analyzers and formal verification frameworks (e.g., Slither, MythX, or Certora) to rigorously validate the contract logic and enhance security prior to any production deployment.

To evaluate core functionality, the *FairChain* prototype was deployed on the Ethereum Goerli testnet. Gas fees for executing smart contracts ranged from ETH 0.002 to 0.005, and transaction confirmation times averaged 12 to 15 s. Under simulated concurrent conditions, the system successfully processed over 100 bio-sample registration transactions without performance degradation or functional errors. These initial results provide early validation of the system's readiness for broader deployment and operational integration.

Environmental Considerations: Given the scale of pharmaceutical manufacturing and regulatory monitoring, energy efficiency is a critical design factor. *FairChain* is deployed on the Ethereum network, which uses a PoS consensus mechanism. This significantly reduces energy consumption compared to earlier PoW models. PoS eliminates the need for high-power mining hardware and minimizes the ecological footprint of transaction validation. Furthermore, *FairChain*'s design deliberately avoids unnecessary on-chain computation, opting instead for off-chain data storage and on-chain hashing to preserve energy efficiency. Future versions may also explore sustainability-focused blockchain networks or layer 2 solutions to further reduce transaction overhead in high-volume deployment scenarios.

7.3. Web Application Implementation

In this part, the main purpose is to create a user interface that is simple and easy to use. The *FairChain* web application was developed using *HTML*, *PHP*, *CSS*, and *JavaScript*.

The *FairChain* website allows different users according to their authorities, which allows them them trace bio-sample information and updates. In addition, it allows different stakeholders to record a DDLC, starting with the registration of a new bio-sample and landowner and ending with obtaining regulatory approval.

7.4. System Integration

System integration is the process of combining sub-systems into a single system in a comprehensive manner and ensuring that each component is working properly. Our website was connected to *phpMyAdmin* database by constructing a connection in our *PHP* files using JavaScript. After that, we added our functions sequentially to ensure that they all worked correctly. The last step was to connect our website with the blockchain via *Web3.js*, which is a collection of libraries allowing for interaction with a local or remote *Ethereum* node using *HTTP*, *IPC*, or *WebSocket*. The process is illustrated in Figure 18.

```
<!-- connecting to the blockchain using web3.js -->
<script src="https://cdn.jsdelivr.net/gh/ethereum/web3.js@1.0.0-beta.34/dist/web3.min.js"></script>
    <script src="https://code.jquery.com/jquery-3.2.1.slim.min.js"></script>
<script>
var web3;
if (typeof web3 !== 'undefined') {
   //if the user has MetaMask installed, then use it
    web3 =window.ethereum;
    ethereum.enable();
    web3= new Web3(web3.currentProvider);
    web3 = new Web3(Web3.givenProvider);
} else {
    //if the user doesn/t have metamask installed , use the private node( locally)
    web3 = new Web3(new Web3.providers.HttpProvider("http://localhost:8545"));
    web3=new Web3(web3.currentProvider);
    //ask the user to install metamask
  window.alert("please download metamask to use this site properly");
}
//////////////////////////////////////

// Entering the contract address to intract with
var address = '0xBDaa1E49c2343aC838b784C2611f9ef206A6b721';
//Entring the contract abi-contract structure-(from remix)
var abi = [{"inputs": [{"internalType": "uint256","name": "","type": "uint256"}],"name": "BioSampleAccts'
    //instatiating the smart contract opject
var fairchainContract = new web3.eth.Contract(abi,address);
</script>
```

Figure 18. *Web3.js* connection.

To maintain synchronization between off-chain and on-chain records, *FairChain* uses a two-phase commit-like approach:

1. The off-chain MySQL database write is initiated first.
2. A cryptographic hash of the stored record is computed.
3. This hash is then submitted in a blockchain transaction using *Web3.js* and stored immutably on-chain.
4. If the on-chain transaction fails, the database write is rolled back to prevent inconsistency.

This mechanism ensures eventual consistency between both storage components. Each on-chain record references its off-chain counterpart through a unique hash, which supports validation and rollback if discrepancies occur. Additionally, *FairChain* maintains synchronization logs that timestamp blockchain submission and confirm transaction hashes, ensuring reliable audit trails.

The *FairChain* website was mainly designed to allow users to send and retrieve data to and from both the database (off-chain data) and the blockchain ledger (on-chain data). The user sends the data while recording the bio-sample information and landowner throughout the entire DDLC process. Receiving the data occurs when the user needs to view the bio-sample ledger (which contains all the information and updates of the bio-sample data that were sent previously). Sending and receiving from the database was completed successfully and worked exactly as designed.

Furthermore, to support interoperability with external clinical research databases such as the WHO International Clinical Trials Registry Platform (ICTRP) [36], *FairChain* is

designed with a modular architecture that includes Application Programming Interfaces (APIs) following the principles of Representational State Transfer (REST) [37] and using standard HTTP methods (GET, POST, PUT, DELETE) to access and manipulate resources (often through URLs). These APIs facilitate two-way communication, allowing *FairChain* to transmit key trial milestones or NDA statuses while also retrieving trial metadata from global registries.

Moreover, to ensure semantic consistency, data mappings align with Clinical Data Interchange Standards Consortium (CDISC) standards and blockchain entries may reference external identifiers such as National Clinical Trial Identifiers. This interoperability layer ensures that *FairChain* can seamlessly interact with global regulatory and research ecosystems, enhancing its practical applicability and regulatory compliance.

The connection between the *FairChain* database and the blockchain ledger is established through *Web3.js*, a JavaScript library that allows for interaction with the Ethereum blockchain from the application backend. When users register or update data (e.g., bio-samples or NDA statuses), the complete records are stored off-chain in a MySQL database managed through PHP. Concurrently, cryptographic hashes or unique record identifiers are generated and transmitted to the blockchain using *Web3.js* functions embedded in the PHP backend. These hashes are stored in the blockchain via Ethereum smart contracts, allowing tamper-proof references to the original off-chain data. This dual-anchoring mechanism ensures that data stored in the off-chain database can be validated through their immutable on-chain counterpart, thereby maintaining synchronization and enabling auditability without compromising performance or privacy.

7.5. Data Consistency and Security Measures

To ensure consistency and security of bio-sample and drug-related data, *FairChain* implements the following mechanisms:

- **Hash Verification:** All off-chain entries include a hash submitted to the blockchain, enabling verification that stored data have not been altered.
- **Role-Based Access Control (RBAC):** Smart contracts and the web application enforce strict role-based permissions, limiting access to sensitive operations.
- **Transactional Integrity:** The system uses error-handling protocols to confirm on-chain recording before finalizing any database changes, which ensures consistent states.
- **Audit Logging:** All modifications are logged with timestamps and actor IDs, enabling accountability and traceability.

This ensures a secure and tamper-resistant platform that aligns with privacy regulations such as GDPR and HIPAA while ensuring consistency between the decentralized and centralized storage environments.

8. Discussion and Conclusions

After several years of COVID-19 vaccination development and distribution, the market for fake drugs and vaccines remains enormous. Furthermore, normal DDLC typically takes a decade, without any means to track the drug origins or record the original landowners' information throughout the process of drug and vaccine development. As a result, DDLC often loses track of the geographical sources of the bio-samples used in drug development and the information of original landowners (loss of originality). Landowners' inability to exercise their right to obtain profits hinders the implementation of the UNDP Nagoya Protocol, which provides a transparent legal framework for effectively implementing one of the three objectives of the Convention on Biological Diversity concerning access to genetic resources and the fair and equitable sharing of emerging benefits [17]. There is currently no digital health solution available for implementation of the UNDP Nagoya Protocol.

Therefore, our project aims to address this gap by developing *FairChain* as the first digital health solution to maintain bio-sample originality and fairly distribute drug development data and updates between drug companies and researchers. *FairChain* accomplishes this in a fast, transparent, and integrated manner by using blockchain technology. The unique features of blockchain technology make it the right choice to implement the UNDP Nagoya Protocol. First, the decentralized structure of blockchain electronic ledgers allows for a shared ledger to be fairly and transparently distributed on thousands of computers around the world. Second, blockchains cannot be altered retrospectively, allowing ownership data to be recorded and transferred without external verification. In this way, *FairChain* represents the first technical implementation of UNDP Nagoya Protocol for trust and fairness throughout the drug development process.

While *FairChain* leverages blockchain for transparency and traceability, we recognize that public blockchain networks face challenges related to scalability, throughput, and storage capacity. To address these, *FairChain* implements a hybrid architecture in which large-scale pharmaceutical and clinical data are stored off-chain in a secure relational database, with only essential metadata and hashed references recorded on-chain. This approach reduces blockchain bloat and ensures system responsiveness. Future work will explore integrating distributed storage systems (e.g., IPFS), adopting enterprise-grade blockchains such as Hyperledger Fabric, and implementing layer 2 scaling solutions to further improve scalability while supporting the real-time access needs of large pharmaceutical companies and regulatory agencies.

8.1. Risk Analysis and Security Considerations

Despite its advantages, deploying blockchains in pharmaceutical ecosystems introduces specific security and operational risks. *FairChain* addresses these proactively through a multi-layered security strategy:

- **Smart Contract Vulnerabilities:** The use of Solidity exposes the system to risks such as re-entrance attacks and integer overflows. To mitigate this, smart contracts were tested using the Remix and Truffle frameworks. Future versions will integrate formal verification tools such as Slither or MythX.
- **Oracle and Middleware Risks:** The integration of off-chain data sources through *Web3.js* may create attack surfaces for oracle manipulation. To minimize this issue, data feeds are authenticated and logs are hash-matched with the blockchain ledger to detect tampering.
- **Private Key Exposure:** Private key theft could compromise system integrity; thus, *FairChain* recommends hardware wallets and multi-signature authorization for sensitive actions.
- **Centralized Database Weaknesses:** MySQL is used to store off-chain data, which could be a single point of failure. Mitigation approaches could include encryption, role-based access control, and future deployment of database replication with clustering.
- **Denial of Service (DoS):** Excessive or malformed transactions can disrupt smart contract functions. To prevent such abuse, *FairChain* introduces gas usage limits and transaction validation mechanisms.

Through this layered approach, *FairChain* balances decentralization with practical risk mitigation to support trustworthy pharmaceutical traceability systems.

8.2. Limitations and Jurisdictional Challenges

While *FairChain* provides traceability and tamper-resistance through blockchain, several limitations persist.

- **Validator Selection in Proof-of-Stake (PoS):** In PoS systems such as Ethereum, validators are chosen based on the quantity of staked tokens. While this promotes economic fairness and discourages malicious behavior, it may concentrate influence among large stakeholders, potentially raising concerns about decentralization. Moreover, validator availability and network liveness are subject to their economic incentives and geographical distribution, which could affect transaction confirmation reliability during high-load events.
- **Security Risks beyond Smart Contracts:** Although the core smart contracts have been tested and validated, broader attack vectors persist. These include endpoint vulnerabilities in *Web3.js* interfaces, client-side attacks such as phishing, and risks involving key management. Decentralized systems are also vulnerable to coordinated attacks (e.g., 51% attacks in smaller PoS networks), although the Ethereum mainnet has high resilience due to its size. Future work will consider integrating Hardware Security Modules (HSMs) and formal verification of contracts.
- **Data Governance across Jurisdictions:** Data sovereignty and privacy regulations vary across countries, affecting how off-chain data must be handled. For example, the European Union's GDPR mandates strict controls over data portability and the right to erasure, while some jurisdictions may require data localization. *FairChain* supports these requirements by keeping personal data off-chain, but cross-border deployments may require compliance mapping and legal risk assessments to ensure lawful interoperability. Addressing these challenges requires regulatory harmonization or middleware solutions that adapt to local data laws.

8.3. Deployment Costs vs. Long-Term Benefits

Initial deployment costs for *FairChain* include smart contract development and auditing, server infrastructure for hosting the off-chain database, and staff training for regulatory institutions and pharmaceutical stakeholders. On public blockchains such as Ethereum, gas fees for registering bio-sample transactions are estimated at ETH 0.002 to 0.005 per transaction; however, these costs are expected to decline with scaling solutions such as layer 2 rollups.

In contrast, long-term benefits include enhanced transparency in the drug development lifecycle, automated enforcement of benefit-sharing obligations under the Nagoya Protocol, tamper-proof traceability for regulators, and reduced compliance costs through auditable digital records. These gains promote institutional trust, data accountability, and greater collaboration across global pharmaceutical research ecosystems. Thus, the investment in blockchain deployment is justified by the systemic improvements in fairness, efficiency, and legal assurance.

Eventually, future work will include conducting a controlled pilot deployment of *FairChain* in collaboration with regulatory or academic institutions. This pilot phase will allow for comprehensive stress testing, scalability benchmarking, and performance evaluations under realistic load conditions. Planned metrics include transaction throughput, latency under high concurrency, and long-term gas cost trends. These insights will guide system optimization and inform future enhancements for large-scale adoption.

Author Contributions: Conceptualization, S.A.; Software, S.A.A. (Shaima A. Alnehmi), A.A.A., A.H.A. and N.A.A.; Formal analysis, A.A.A., A.H.A. and N.A.A.; Investigation, K.A. and H.A.S.; Data curation, K.A.; Writing—original draft, S.A.A. (Shaima A. Alnehmi) and S.S.A.; Writing—review & editing, S.A. and H.A.A.; Visualization, S.S.A.; Supervision, S.A. and H.A.A.; Project administration, S.A.A. (Sara A. Alsalamah); Funding acquisition, S.A. and H.A.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: Data are contained within the article.

Acknowledgments: The authors extend their sincere gratitude to the subject matter experts, Sara Al-Rashood and Abdulmohsen Abanumai for their valuable time and insightful contributions during the interviews conducted for this study. Their contributions have been instrumental in shaping the findings of this research.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Lichtenberg, F.R. *Pharmaceutical Innovation, Mortality Reduction, and Economic Growth*; Working Paper Series; National Bureau of Economic Research: Cambridge, MA, USA, 1998; Volume 6569.
2. Buxbaum, J.D.; Chernew, M.E.; Fendrick, A.M.; Cutler, D.M. Contributions of Public Health, Pharmaceuticals, and Other Medical Care to US Life Expectancy Changes, 1990–2015. *Health Aff.* **2020**, *39*, 1546–1556. [CrossRef] [PubMed]
3. Strovel, J.; Sittampalam, S.; Coussens, N.P.; Hughes, M.; Inglese, J.; Kurtz, A.; Andalibi, A.; Patton, L.; Austin, C.; Baltezor, M.; et al. Early Drug Discovery and Development Guidelines: For Academic Researchers, Collaborators, and Start-up Companies. In *Assay Guidance Manual*; Markossian, S., Grossman, A., Baskir, H., Arkin, M., Auld, D., Austin, C., Baell, J., Brimacombe, K., Chung, T.D.Y., Coussens, N.P., et al., Eds.; Eli Lilly & Company and the National Center for Advancing Translational Sciences: Bethesda, MD, USA, 2016.
4. Balcells-Camps, M. (Principal Research Scientist, MIT Institute for Medical Engineering and Science (IMES), Massachusetts Institute of Technology (MIT), Cambridge, MA, USA). Personal communication, 2020.
5. U.S. Food and Drug Administration. The Drug Development Process. Available online: https://www.fda.gov/patients/learn-about-drug-and-device-approvals/drug-development-process (accessed on 25 February 2020).
6. McCreath, S.B.; Delgoda, R. *Pharmacognosy: Fundamentals, Applications and Strategies*; Academic Press: Cambridge, MA, USA, 2017.
7. Renner, S.C.; Neumann, D.; Burkart, M.; Feit, U.; Groger, A.; Paulsh, A.; Paulsh, C.; Sterz, M.; Vohland, K. Import and export of biological samples from tropical countries–considerations and guidelines for research teams. *Org. Divers. Evol.* **2012**, *12*, 81–98. [CrossRef]
8. Lipsky, M.S.; Sharp, L.K. From idea to market: The drug approval process. *J. Am. Board Fam. Pract.* **2001**, *14*, 362–367.
9. Convention on Biological Diversity. About the Nagoya Protocol. Available online: https://www.cbd.int/abs/about/ (accessed on 14 October 2024).
10. Presutti, L. Blockchain Technology: Challenges and Legal Issues. CyberLaws. 2018. Available online: https://www.cyberlaws.it/2018/blockchain-technology-challenges-legal-issues/ (accessed on 14 October 2024).
11. Blockchain.com. Blockchain.com Privacy Policy. 2022. Available online: https://www.blockchain.com/legal/privacy (accessed on 14 October 2024).
12. Pacific BioLabs. Stages of Drug Development-Pacific Biolabs. 2020. Available online: https://pacificbiolabs.com/stages-of-drug-development (accessed on 16 October 2024).
13. U.S. Food and Drug Administration. Step 2: Preclinical Research. 2020. Available online: https://www.fda.gov/patients/drug-development-process/step-2-preclinical-research (accessed on 16 October 2024).
14. U.S. Food and Drug Administration. Step 3: Clinical Research. 2020. Available online: https://www.fda.gov/patients/drug-development-process/step-3-clinical-research (accessed on 16 October 2024).
15. U.S. Food and Drug Administration. Step 4: FDA Drug Review. 2020. Available online: https://www.fda.gov/patients/drug-development-process/step-4-fda-drug-review (accessed on 16 October 2024).
16. U.S. Food and Drug Administration. Step 5: FDA Post-Market Drug Safety Monitoring. 2020. Available online: https://www.fda.gov/patients/drug-development-process/step-5-fda-post-market-drug-safety-monitoring (accessed on 16 October 2024).
17. International Union for Conservation of Nature and Natural Resources (IUCN). The Nagoya Protocol. Available online: https://www.iucn.org/theme/global-policy/our-work/convention-biological-diversity-cbd/nagoya-protocol (accessed on 16 October 2024).
18. What Is Blockchain Technology? A Step-By-Step Guide for Beginners. Available online: https://www.iucn.org/theme/global-policy/our-work/convention-biological-diversity-cbd/nagoya-protocol (accessed on 16 October 2024).
19. Itemtracker-Sample Management & Tracking Software. Available online: https://www.itemtracker.com/ (accessed on 18 October 2024).
20. Laboratory Information Management Software (LIMS) LIMS Software. Available online: https://www.autoscribeinformatics.com/lims-laboratory-information-management-system (accessed on 18 October 2024).
21. Brooks Life Sciences Brooks Life Sciences. Available online: https://www.brookslifesciences.com/ (accessed on 18 October 2024).
22. Sustainable Sample Tracking Fluics Connect. Available online: https://fluics.com/ (accessed on 18 October 2024).

23. Yazdinejad, A.; Dehghantanha, A.; Karimipour, H.; Srivastava, G.; Parizi, R.M. A robust privacy-preserving federated learning model against model poisoning attacks. *IEEE Trans. Inf. Forensics Secur.* **2024**, *19*, 6693–6708. [CrossRef]
24. Mettler, M. Blockchain technology in healthcare: The revolution starts here. In Proceedings of the 2016 IEEE 18th International Conference on e-Health Networking, Applications and Services (Healthcom), Munich, Germany, 14–17 September 2016; pp. 1–3.
25. Yue, X.; Wang, H.; Jin, D.; Li, M.; Jiang, W. Healthcare data gateways: Found healthcare intelligence on blockchain with novel privacy risk control. *J. Med. Syst.* **2016**, *40*, 218. [CrossRef] [PubMed]
26. Benchoufi, M.; Ravaud, P. Blockchain technology for improving clinical research quality. *Trials* **2017**, *18*, 335. [CrossRef] [PubMed]
27. Azaria, A.; Ekblaw, A.; Vieira, T.; Lippman, A. MedRec: Using blockchain for medical data access and permission management. In Proceedings of the 2016 IEEE Open and Big Data Conference, Washington, DC, USA, 5–8 December 2016; pp. 25–30.
28. Hylock, R.H.; Zeng, X. A review and evaluation of blockchain systems for healthcare applications. *J. Biomed. Inform.* **2022**, *127*, 104005.
29. MediLedger. MediLedger Network: Secure Pharmaceutical Supply Chains. Available online: https://www.mediledger.com (accessed on 16 October 2024).
30. IBM Blockchain. IBM Pharma Ledger. Available online: https://www.ibm.com/blockchain/solutions/pharmaceutical (accessed on 16 October 2024).
31. VeChain. VeChain ToolChain™. Available online: https://www.vechain.org (accessed on 16 October 2024).
32. PharmaLedger EU. PharmaLedger Blockchain Initiative. Available online: https://www.ihi.europa.eu/projects-results/project-factsheets/pharmaledger (accessed on 16 October 2024).
33. BlockPharma. Blockchain Against Drug Counterfeiting. Available online: https://www.blockchainalmanac.com/supply-chain-and-logistics/blockpharma/ (accessed on 16 October 2024).
34. Hyperledger Foundation. Hyperledger Fabric for Pharma. Available online: https://www.hyperledger.org (accessed on 16 October 2024).
35. Phpmyadmin. Available online: https://www.phpmyadmin.net/ (accessed on 24 October 2024).
36. World Health Organization (WHO). International Clinical Trials Registry Platform (ICTRP). Available online: https://www.who.int/clinical-trials-registry-platform (accessed on 24 October 2024).
37. Fielding, R.T. Architectural Styles and the Design of Network-based Software Architectures. Ph.D. Thesis, University of California, Irvine, CA, USA, 2000.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.